

# Assignment 1: From Dirty Data to Predictive Models

## Submission

- A runnable Jupyter/Colab Notebook (.ipynb).
- A PDF report (10-12 pages).
  - On page 1, include the Colab link and a page map: which pages cover Step 1, 2, 3, 4, 5, and the AI Disclosure.
  - Note: The report is for analysis and insights. Do NOT copy-paste code blocks into the main body of the PDF.

## A Note Before You Start

Use AI tools freely on this assignment. That is not a loophole, it is how the job works now. But there is one thing AI cannot do for you: **understand your own code**.

Before you submit, read your notebook top to bottom and explain each block to yourself, out loud if that helps. Why this imputation? Why this alpha? What would break if you removed this line? If you cannot answer, you do not own that code yet, and an interviewer will find out in about ninety seconds. I have watched it happen to strong students. I would rather it happen here, where the cost is a few extra hours, than in the room where the offer is decided.

The report is where you show us you understand. We read it carefully, and we are rooting for you.

## Objective

The goal of this assignment is to guide you through the process of building an end-to-end flow to turn data to predictive models, using what you have learned so far. You will:

- Clean and transform a real dataset that contains missing values, noisy data, and categorical variables.
- Engineer meaningful features to improve model performance.
- Train and compare 2 different modeling approaches:
  - Generative: Naive Bayes
  - Discriminative: Linear Regression
- Evaluate your models using appropriate metrics and visualizations.
- Reflect on your workflow and disclose any AI tools used.

By the end of this assignment, you shall have hands-on experience with data preprocessing, feature engineering, model training, basic evaluation, and interpretation.

## Datasets

Choose one of the following datasets. Both datasets contain real-world challenges such as missing values and categorical variables.

## Titanic Survival Dataset

Task: Predict whether a passenger survived or not.

Dataset link: [Kaggle Titanic Dataset](#)

Notes: Requires a Kaggle account.

## Heart Disease Prediction Dataset

Task: Predict whether a patient has heart disease based on medical attributes.

Dataset link: [UCI Heart Disease Dataset](#)

## Steps

To resolve this supervised learning task, your assignment will consist of the following steps:

1. Data Cleaning
  - a. Handle missing values (imputation, dropping, or flagging).
  - b. Address noisy or inconsistent values.
  - c. Justify your choices (why impute vs. drop?).
2. Feature Engineering
  - a. Apply appropriate transformations (normalization, standardization, log-scaling).
  - b. Encode categorical variables (one-hot encoding or other methods).
  - c. Optionally, construct new features (e.g., ratios, group statistics).
3. Model Training
  - a. Train a Naive Bayes classifier (BernoulliNB or GaussianNB depending on your dataset), and make sure to apply Laplace smoothing (add- $\alpha$ ).
    - i. You should experiment with at least 2 different values of alpha (e.g., 1.0 vs. 0.01) and compare the results.
  - b. Train a Linear Regression model.
    - i. Although Linear Regression is typically used for **regression tasks**, in this assignment we will also apply it to a binary classification problem with a probability threshold as 0.5.
    - ii. Reminder: To prevent overfitting and to explore the effect of regularization, you are encouraged to also try:
      1. Ridge Regression (L2 regularization)
      2. LASSO Regression (L1 regularization)
  - c. Ensure fair comparison (same train/test split).
4. Model Evaluation
  - a. [Required] Accuracy + Confusion Matrix. [Encouraged] Precision, Recall, F1-score. [Bonus] ROC + AUC
  - b. Include at least one visualization (confusion matrix heatmap, ROC curve).
  - c. Explain what happened when you used smoothing vs. without smoothing.
5. Report Writing (10-12 pages)

- a. Introduction: problem definition + dataset description.
- b. Data Cleaning: your steps, reasoning, and before/after examples.
- c. Feature Engineering: transformations and encodings applied.
- d. Model Comparison: training setup, evaluation results.
- e. Discussion: interpretation of results and limitations.
- f. **AI Tool Usage Disclosure:**
  - i. List AI tools used (e.g., ChatGPT, Gemini, Claude).
  - ii. Describe what these tools contributed (e.g., generating starter code, visualization, debugging).
  - iii. **Clarify which parts were your own contribution.**

## Pointers & Hints

To help you get started as the 1st assignment:

- Libraries: `scikit-learn`, `pandas`, `numpy`, `matplotlib/seaborn`.
- Functions:
  - `train_test_split` for splitting data.
  - `classification_report`, `confusion_matrix` for evaluation.
  - `cross_val_score` for validation (optional).
- Models:
  - `sklearn.naive_bayes.BernoulliNB`
  - `sklearn.naive_bayes.GaussianNB`
  - `sklearn.linear_model.LinearRegression`
- Visualization:
  - Use `seaborn.heatmap` for confusion matrix.
  - [Optional] Use `sklearn.metrics.roc_curve` and `auc` for ROC plots.

## Report Formatting & Visualization Rules:

- **Visuals:** All plots, confusion matrices, and figures must be embedded directly in the report (e.g., take a screenshot of your graph from the notebook and paste it here).
  - We will not open your notebook to see the graphs.
- **Code:** No code in the report, body or appendix. All code goes in the .ipynb you upload. Put the Colab link on page 1 of the report.

## Grading Rubric

| Category                       | Percentage | Description                                                                                    |
|--------------------------------|------------|------------------------------------------------------------------------------------------------|
| Data Cleaning & Transformation | 20         | Clear handling of missing, noisy, and inconsistent data with justification.                    |
| Feature Engineering            | 20         | Correct application of transformations and encodings; creativity in constructing new features. |
| Model Implementation           | 20         | Properly trained Naive Bayes and Linear Regression; fair comparison setup.                     |

|                                        |    |                                                                                                                          |
|----------------------------------------|----|--------------------------------------------------------------------------------------------------------------------------|
| <b>Evaluation &amp; Visualization</b>  | 15 | Correct use of metrics; meaningful visualizations included. Graphs must be visible in the PDF report, not just the code. |
| <b>Discussion &amp; Interpretation</b> | 15 | Thoughtful analysis of results, strengths/limitations, and clear explanations.                                           |
| <b>AI Tool Usage Disclosure</b>        | 10 | Transparent description of AI tool usage and personal contributions.                                                     |